ALWIN T R

Embedded Systems & Robotics Engineer

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PROFILE SUMMARY

Proactive Embedded Systems & Robotics Engineer with hands-on experience in ROS 2 autonomous navigation, real-time microkernel firmware (FreeRTOS / Bare-Metal C++), and industrial sensor systems. Proven track record across STM32, ESP32, and PIC microcontrollers, multi-axis kinematics, radar signal processing (24GHz FMCW), and hardware communications (UART, I2C, SPI, CAN). Experienced in driving embedded products from schematic capture to deployment.

TECHNICAL SKILLS

Programming: C, C++, Embedded C (C++17), Python, Bash, PIC Assembly / XC8
Robotics & Kinematics: ROS 2 (Humble), micro-ROS, Nav2, 2D SLAM (Cartographer/AMCL), Inverse Kinematics, Gazebo, RViz2
Microcontrollers & Hardware: STM32 (ARM Cortex-M), ESP32 Dual-Core, PIC16F877A, ATmega328P, Raspberry Pi 4, PCA9685
Embedded Systems & RTOS: FreeRTOS (Dual-Core Tasks, Queues, Semaphores), Interrupts (ISR), Timers, PWM, ADC, Watchdog
Protocols & IoT: UART, I2C (Fast Mode 400kHz), SPI, CAN Bus, ESP-NOW, MQTT, Wi-Fi 802.11, BLE, HTTP/REST
Tools & Platforms: STM32CubeIDE, MPLAB X, Arduino IDE, Proteus, Git, GitHub, Ubuntu Linux

EXPERIENCE

Embedded Systems & Robotics Engineer — *Mepits Infotech Pvt. Ltd., Kochi* **May 2026 – Present**

- Architecting and deploying embedded firmware and robotics solutions across STM32 and ESP32 microcontrollers.
- Designing modular device drivers and real-time control algorithms with FreeRTOS task synchronization for automation products.
- Conducting board bring-up, peripheral validation, and automated hardware-in-the-loop (HIL) testing, achieving a 99.4% test pass rate.

Robotics Trainee — *Evolve Robotics, Kochi* **Aug 2025 – Apr 2026**

- Architected custom ROS 2 (Humble) nodes in C++ and Python for differential-drive mobile robots, integrating Nav2 stacks for autonomous navigation and dynamic obstacle avoidance.
- Configured 2D SLAM mapping and sensor fusion pipeline fusing LiDAR and wheel odometry, improving localization accuracy in unstructured indoor environments.
- Implemented bare-metal and RTOS-based motor control algorithms, managing PWM signal generation and encoder feedback under deterministic timing constraints.
- Configured and verified embedded bus communication drivers (UART, I2C, SPI) to stream high-frequency telemetry, reducing sensor latency by 35%.

Embedded Systems Intern — *Aester India Pvt. Ltd., Kochi* **Jun 2023 – Jul 2023**

- Engineered embedded C/C++ firmware on AVR and PIC microcontrollers for industrial automation modules.
- Interfaced analog/digital sensors and motor driver circuits, implementing PWM speed regulation and directional control.
- Executed hardware prototyping, PCB breadboard routing, circuit testing, and hardware-in-the-loop debugging using test equipment.

IoT Intern — *Keltron, Thiruvananthapuram* **Jan 2023 – Feb 2023**

- Developed IoT sensor-monitoring prototypes using ESP32 microcontrollers, publishing real-time telemetry via MQTT protocol.
- Interfaced RFID readers, environmental sensors, and relay actuators to build automated access and monitoring systems.
- Conducted peripheral pinout allocation, ADC calibration, and power optimization in Arduino IDE with extensive hardware bench testing.

TECHNICAL PROJECTS

ESP32 6-DOF Bipedal Walking Robot with FreeRTOS Kinematics **ESP32, FreeRTOS, PCA9685, I2C, C/C++**

- Engineered a 6-DOF humanoid bipedal walking robot using dual-core ESP32 with thread isolation (50Hz locomotion task on Core 1).
- Implemented closed-form 2D Inverse Kinematics, quintic zero-jerk swing trajectories, and 12-bit PCA9685 PWM control over 400kHz I2C for 6 high-torque metal-gear servos. Built real-time asynchronous Web Telemetry Studio on Core 0.

24GHz mmWave Human Presence & Micro-Motion Radar Monitor **ESP32, HLK-LD2410C FMCW, UART, C++**

- Developed a sub-millimeter human presence and respiration detection monitor utilizing Hi-Link 24GHz FMCW radar and ESP32.
- Implemented non-blocking UART frame protocol parser (115200 baud) extracting motion energy and static micro-movements across 8 spatial range gates (up to 6m) with 99.2% detection reliability.

Autonomous Reconnaissance Rover with Gas and Metal Detection **ESP32-CAM, Arduino, MQ-2, L298N, C++**

- Built an all-terrain tele-operated surveillance rover with real-time wireless video transmission over local Wi-Fi UDP stream.
- Integrated MQ-2 combustible gas sensor and inductive metal detector with automated threshold alert interrupts and 4WD dual H-bridge motor actuation.

EDUCATION

B.E. in Robotics and Automation — *Dhanalakshmi Srinivasan Engineering College, Perambalur, Tamil Nadu* **2021 – 2025**

Higher Secondary (Computer Science) — *Santa Cruz HSS, Fort Kochi, Ernakulam* **2019 – 2021**